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Influence of different carbon and nitrogen sources on the growth and sclerotia formation of *Sclerotinia sclerotiorum* (Lib.) de Bary, the causal organism of stem rot of Chickpea

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Abstract : Carbon and nitrogen are the most essential elements to be supplemented in the media upon which growth of fungal mycelial and sclerotia formation depends. Without the addition of these macroelements, there was poor mycelial growth and sclerotia formation of the pathogen *Sclerotinia sclerotiorum* (Lib.) de Bary which causes stem rot disease of chickpea. Attack of this pathogen ultimately leads to the decrease in the yield of the crop. In this observation, eight different carbon sources such as sucrose, xylose, galactose, glucose, raffinose, rhamnose, fructose, mellibiose were taken. Among which Sucrose followed by xylose and glucose was the best carbon source for mycelial growth and sclerotia formation of the pathogen. Among the nitrogen sources used for observation, sodium nitrate followed by ammonium sulphate , ammonium nitrate and calcium nitrate was proved to be best inorganic sources for the growth and sclerotia formation of the pathogen. Similarly, among the different organic nitrogen sources used asparagine followed by glycine showed the best sclerotia formation and mycelial growth.

Keywords : *Sclerotinia sclerotiorum*, carbon sources, nitrogen sources, mycelia, sclerotia, Czapek's media.

INTRODUCTION

Stem rot which is also known as sclerotia wilt or white mold, caused by *Sclerotinia sclerotiorum* is a serious disease of chickpea. It infects all the economically important food and feed legumes[1,2]. Carbon and nitrogen both are essential components of the culture media which provides nutrition for the growth and sclerotial formation of the fungus. Nutrients are the factors which provides energy for biosynthesis and thus helps in the growth and development of any microorganism[3,4]. The growth and development of entomopathogenic fungi depends upon the nutrient factors[4,5]. The knowledge of nutritional requirements is the main need in the cultivation of microorganisms. The carbohydrates, proteins, lipids, nucleic acids, are made up of macro elements like carbon, hydrogen, nitrogen, sulphur, phosphorous which are involved in the host pathogen interaction and self defense mechanisms. The preparation used for growth, storage and transport of microorganisms can be used in solid[5] and liquid form[7]. The effect of nutrient sources on the growth and development of microorganisms is studied[4,8]. The fungus which is used and the components of media used for the growth of mycelia as well as spore yield[4]. Nutrients are substances used in biosynthesis and energy release and therefore serve as cardinal impetus towards the viability, survival and sustenance of any organism[3,9] . Nutrient source is an integral determinant of growing and virulence of phytopathogenic fungi[9]. The fungal symbionts are able to transfer carbon (C^{14}), nitrogen (N^{15}), phosphorous (P^{32}), sodium (Na^{22}), and calcium (Ca^{45}) from the nutrient solution into the plant in larger quantities as compared to the non- mycorrhizal roots[10]. Different forms of carbon sources mono-, di-, tri- or polysaccharides evoke different fungal response due to improvement of the specific enzymes for their breakdown and uptake as revealed by the growth and sporulation of the concerned fungus[11]. Nitrogen is an essential element of fungal structure and life processes. Earlier workers reported that

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amino acids were more favorable as a nitrogen source than nitrates or ammonium[9,12]. So, the present work was done upon the different carbon and nitrogen sources to see the effect on the mycelial growth and sclerotia formation of the pathogen *S. sclerotiorum.*

MATERIALS AND METHODS

The isolated pathogen was maintained in Potato Dextrose Agar solid media. Further experiment to see the effect of different carbon and nitrogen sources on the mycelial growth and sclerotia formation of the fungus was done with Czapek's dox media which was proved to be the best liquid media after the potato dextrose media, as this media contain the known amount of carbon and nitrogen sources. Eight different carbon sources such as glucose, sucrose, fructose, mellibiose, xylose, rhamnose, raffinose, galactose were taken. Along with different carbon sources control was also maintained which was without carbon source for the comparison of the mycelial growth and sclerotia formation.50 ml of Czapek's media were taken in 250 ml conical flask and equal amount of carbon was added in different conical flask. The amount of carbon was equal to the amount of carbon present in medium, this was done by calculating their molecular weight and number of carbon present in sugar of Czapek's media i.e. Sucrose. The pathogen was inoculated in all flask containing different carbon sources and were incubated at 30°C for the period of 15, 30, 45 days. The nitrogen source of the media was replaced by nine different inorganic and organic nitrogen sources. Equal amount of nitrogen equivalent to the nitrogen of the media was added in the different conical flask containing Czapek's media. The organic nitrogen sources taken were asparagine, glycine, leucine, 2-thiobarbitosure. And in inorganic sources considered were KNO_3, $NaNO_3$, $Ca(NO_3)$, NH_4NO_3, $(NH_4)_2SO_4$. One control was also maintained which was without nitrogen sources for comparison of the mycelial growth and sclerotia formation of the pathogen. The pathogen were inoculated and incubated at 30°C for the period of 15, 30, 45 days. The experiment was done in three replicates. After the days of incubation period, the mycelial weight of the fungus were measured and number of sclerotia formation were counted. In both cases of carbon and nitrogen sources the fungal mat was filtered with the help of Whatsmann filter paper and the filter paper containing mycelial mat was dried in hot air oven.

Dry weight of mycelium = Weight of dried filter paper with fungal mat - Weight of the filter paper.

Table1. Effect of different carbon sources for the mycelial growth and sclerotia formation.

Different Carbon Sources	Incubation period in days	Dry wt. of mycelia	No. of Sclerotia
Glucose	15 days	1.461 mg	05
	30 days	2.382mg	13
	45 days	2.682 mg	10
Sucrose	15 days	1.565 mg	03
	30 days	3.385 mg	10
	45 days	3.875 mg	12
Xylose	15 days	1.153 mg	00
	30 days	2.895 mg	15
	45 days	3.228 mg	20
Fructose	15 days	0.540 mg	15
	30 days	1.390 mg	08
	45 days	1.640 mg	06
Mellibiose	15 days	0.432 mg	00
	30 days	0.932 mg	07
	45 days	1.082 mg	09

Different Carbon Sources	Incubation period in days	Dry wt. of mycelia	No. of Sclerotia
Galactose	15 days	0.250 mg	00
	30 days	0.721 mg	05
	45 days	0.896 mg	08
Raffinose	15 days	0.229 mg	00
	30 days	1.029 mg	06
	45 days	1.265 mg	08
Rhamnose	15 days	0.328 mg	00
	30 days	0.760 mg	09
	45 days	0.990mg	12
Control	15 days	0.029 mg	00
	30 days	0.091 mg	00
	45 days	0.111 mg	00

From the above table 1, it was evident that among the media containing different carbon sources, sucrose followed by xylose and glucose showed maximum mycelial growth, and xylose followed by fructose, glucose and sucrose were found to be best for sclerotia formation by the pathogen. The poor mycelial growth and poor sclerotia formation of the pathogen was found in control. From the experiment, it is evident that for the better mycelial growth and sclerotia formation, the carbon source are important and essential supplement of the medium.

Table 2. Effect of organic and inorganic nitrogen sources for the mycelial growth and sclerotia formation.

Different nitrogen Sources	Incubation period in days	Dry wt. of mycelia in mg	No. of Sclerotia
Asparagine	15 days	2.188 mg	12
	30 days	3.218 mg	20
	45 days	4.020mg	10
Glycine	15 days	1.065 mg	10
	30 days	1.995 mg	15
	45 days	2.558 mg	20
Leucine	15 days	0.893 mg	03
	30 days	1.593 mg	06
	45 days	2.056 mg	09
2-Thiobarbitosure	15 days	0.250 mg	00
	30 days	0.450 mg	00
	45 days	0.550mg	02
Ammonium Sulphate	15 days	1.582 mg	04
	30 days	2.832 mg	10
	45 days	3.707 mg	30
Ammonium nitrate	15 days	1.226 mg	02
	30 days	2.176 mg	05
	45 days	2.908 mg	12

Different nitrogen Sources	Incubation period in days	Dry wt. of mycelia in mg	No. of Sclerotia
Calcium nitrate	15 days	1.238 mg	06
	30 days	2.038 mg	07
	45 days	2.560 mg	15
Potassium nitrate	15 days	0.761 mg	00
	30 days	1.391 mg	04
	45 days	1.841 mg	05
Sodium nitrate	15 days	1.642 mg	03
	30 days	3.227 mg	11
	45 days	4.127 mg	10
Control	15 days	0.321 mg	00
	30 days	0.601 mg	00
	45 days	0.751 mg	00

From the above table 2, it was evident that media containing different inorganic nitrogen sources, maximum mycelial growth was noticed that Sodium nitrate followed by ammonium sulphate and ammonium nitrate, whereas calcium nitrate and potassium nitrate showed the moderate mycelial growth. But in case of organic sources, the maximum growth of the mycelium and sclerotia formation was observed in asparagine followed by glycine. Moderate growth was found in leucine and fair growth and less sclerotia formation was observed in 2- thiobarbitosure. Poor growth and no sclerotia formation was recorded in control. The findings of the observation support that nitrogen sources in the medium was essential component for the mycelial growth and sclerotia formation of the fungus.

RESULTS AND DISCUSSION

Carbon sources are important supplement for the media which provides nutritional requirements of the pathogen. Among eight different carbon sources , Sucrose was found to be the best for the mycelial growth. It was followed by the xylose and glucose. Similar to our findings, sucrose or glucose as a carbon sources was found to be best for the maximum mycelial yield of *T. harzianum*[13]. Similarly, among the different carbon sources, sucrose and glucose were the best for sporulation of *T. koningii* and *T. harzianum*[14]. In our finding maximum sclerotial formation was seen in xylose and fructose but sucrose and glucose also showed good sclerotial production. Similar to our findings, dextrose and sucrose was regarded as better carbon sources for the growth of *Colletotrichum glaeosporiodes*[15]. Similar results were observed for the mycorrhiza of forest plants where the effect of various sugars were seen and was noticed that the maximum mycelial growth recorded glucose, sucrose, maltose[16]. Similar to our findings glucose and sucrose are to the best carbon sources observed for the growth and sporulation of *Curvalaria clavata*[17] . Some more findings were observed where the medium containing L-xylose showed the best growth and sporulation of *Helminthosporum maydis*[18]. In our findings, also xylose was found to be second best among carbon sources for mycelial growth and maximum sclerotia formation. Contradictory to our findings, sucrose containing medium was showed poor growth and sporulation of *Septoria tritici*[19]. In our work, sucrose was the best for the mycelial growth of the pathogen. Contradictory to our findings fructose showed maximum colony growth formation in Trichoderma strain (T_{36})[20]. Similar to our findings glucose showed the maximum mycelial growth followed by fructose for growth of *Rhizopus* species[21]. Similarly, more findings are present in which fructose supported growth next to glucose[22] . Confirmatory to our findings fructose after glucose was found to be the best carbon source for growth of *Rhizopus* species[23].

Organic and inorganic nitrogen sources in the media are necessary for the mycelial growth and sclerotia formation of the pathogen. Nitrogen sources also provide nutrition for the growth of the pathogen. Asparagine followed by glycine enhances the mycelial growth and sclerotia formation among different organic sources. Like the findings of some earlier worker in our finding there was good mycelial with asparagine used as nitrogen source as compared to nitrate or ammonium[9,12].In case of

inorganic sources sodium nitrate followed by ammonium sulphate, ammonium nitrate was found to be the best source for excellent mycelial growth. The best sclerotia formation was observed in ammonium sulphate followed by sodium nitrate and ammonium nitrate. Similar to our findings, maximum growth and sporulation of T.harzianum was found that sodium nitrate, ammonium sulphate and ammonium nitrate[24,25]. Similar to our findings the maximum growth and sporulation of T.viride were observed by ammonium form of nitrogen as compared to nitrite or nitrate forms[22,25,26]. Like our findings, T.viride isolates grew best in media having nitrogen sources as asparagines and ammonium nitrate[27]. KNO_3 followed by $(NH_4)_2SO_4$, NH_4NO_3 showed highest growth for production of *Beauveria bassiana*[28]. This is contradictory to our findings as KNO_3 showed the moderate mycelial growth in our results. KNO_3 showed the maximum growth of C. gloeosporioides[29] which is contradictory to our findings for the growth and sporulation of the pathogen. As in our findings KNO_3 showed moderate mycelia growth. The next best growth was found in sodium nitrate, ammonium sulphate and peptone, while calcium nitrate showed moderate growth of the pathogen[29]. This finding is similar to our observation. Among organic nitrogen source, the fungus produced maximum biomass with L-arginine and L-asparagine to a fair extent for the growth of *Curvularia pallescens*[30], *Sclerotiorum cepivorum*[31], *Hirsutella rhossiliensis*[32], *(Pestalotiopsis submerses, Flagellospora penicilloides)*[33]. This result is contradictory to our findings, as in our observation among the organic sources asparagines followed by glycine showed the highest mycelial growth and sclerotia formation by the pathogen. Ammonium sulphate and peptide were the best nitrogen sources for the growth of *T. harzianum*, but urea was the worst among different nitrogen sources[23,34] . This study was similar to our findings in case of inorganic sources.

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